

# Week 07: Building and checking RAG

Generative AI | MSc teaching notes and further reading | 2027

## Learning target

Connect retrieved evidence to an answer and abstain when support is absent.

## Context and teaching plan

The college-policy assistant answers questions with policy IDs.

| Contact time        | Suggested use                                                                                                    |
|---------------------|------------------------------------------------------------------------------------------------------------------|
| Two hours: concepts | Case introduction, four ideas, worked example and misconception.                                                 |
| Two hours: practice | 10 min recall, 20 min demo, 40 min guided practice, 25 min independent change, 15 min review, 10 min exit check. |

## Explain the key ideas

### RAG

Retrieve relevant text for the current request, then draft an answer bounded by that text.

Teacher question: Where does rag appear in the weekly case, and what could go wrong if we omit it?

### Citation support

Check every material claim against the cited passage.

Teacher question: Where does citation support appear in the weekly case, and what could go wrong if we omit it?

### No-answer case

If approved sources lack an answer, return a clear unsupported status.

Teacher question: Where does no-answer case appear in the weekly case, and what could go wrong if we omit it?

### Test split

Use 20 practice questions for improvements and keep ten separate questions for final checking.

Teacher question: Where does test split appear in the weekly case, and what could go wrong if we omit it?

## Worked example and answer guide

- Input: P3: Students may request one deadline extension within seven days.
- Change or question: Q: How many extensions are allowed?
- Supported observation: A: One deadline extension [P3].

Reasoning: state the given evidence first, then the calculation or rule, then the conclusion. If a value is absent, say what cannot be concluded. Keep any simulated mechanism distinct from a trained model or deployed service.

## Teach the week as a story

The cited deadline answer: The assistant cites P3 for “two extensions,” but P3 permits one. A citation is present. Is the answer supported?

Ask for a quick prediction before each technical explanation. Return to the prediction after the example and ask what evidence changed the answer. The analogies below are memory aids, and each has a stated limit.

### RAG: analogy, mechanism and check

Familiar picture: An open-book examiner finds the page before writing the answer.

| Familiar part | Technical meaning |
|---------------|-------------------|
| Question      | query             |
| Open page     | retrieved passage |
| Answer        | cited reply       |

Where this analogy breaks: Retrieval can pick an irrelevant page; answer quality depends on what was found.

Actual idea: Retrieve relevant text for the current request, then draft an answer bounded by that text.

Work it aloud: P3 states one extension in seven days; answer “one [P3]” after retrieving P3.

Ask before revealing: Which step happens before drafting?

Supported response: Find the relevant allowed passage.

Look for this misconception: A citation added afterward repairs a false answer.

### Citation support: analogy, mechanism and check

Familiar picture: A footnote is useful only when its source actually says the claim.

| Familiar part | Technical meaning |
|---------------|-------------------|
| Sentence      | claim             |
| Footnote      | source ID         |

|                  |                        |
|------------------|------------------------|
| <b>Highlight</b> | <b>supporting span</b> |
|------------------|------------------------|

Where this analogy breaks: Checking entailment can involve nuance; use a human for disputed cases.

Actual idea: Check every material claim against the cited passage.

Work it aloud: “Two extensions [P3]” is unsupported because P3 says one, even though the ID exists.

Ask before revealing: What must be checked beyond the presence of a citation?

Supported response: The passage must support the exact claim.

Look for this misconception: Any source ID proves the sentence.

## No-answer case: analogy, mechanism and check

Familiar picture: A careful researcher leaves a blank when the archive lacks evidence.

|                      |                          |
|----------------------|--------------------------|
| <b>Familiar part</b> | <b>Technical meaning</b> |
| <b>Missing page</b>  | no relevant passage      |
| <b>Blank</b>         | abstain                  |
| <b>Search log</b>    | reason                   |

Where this analogy breaks: A retrieval miss may hide a real answer elsewhere; report the search limit.

Actual idea: If approved sources lack an answer, return a clear unsupported status.

Work it aloud: Ask about parking permits; none of five policies covers it, so say evidence is unavailable.

Ask before revealing: What should the system say?

Supported response: It lacks a relevant approved source in this corpus.

Look for this misconception: The model should guess from common knowledge.

## Test split: analogy, mechanism and check

Familiar picture: A practice quiz guides choices; an unopened exam checks the choice.

|                         |                          |
|-------------------------|--------------------------|
| <b>Familiar part</b>    | <b>Technical meaning</b> |
| <b>20 questions</b>     | development set          |
| <b>Search setting</b>   | chosen configuration     |
| <b>10 new questions</b> | final check              |

Where this analogy breaks: Ten final cases are still a small sample; report each failure and denominator.

Actual idea: Use 20 practice questions for improvements and keep ten separate questions for final checking.

Work it aloud: Compare top-k 1 and 2 on 20 practice questions, freeze the choice, then open ten final cases.

Ask before revealing: Why not tune again after seeing the ten final answers?

Supported response: That would contaminate the independent check.

Look for this misconception: Reusing final questions improves an unbiased estimate.

## New situation for discussion

A fresh question asks about a rule absent from all five policies. What should the assistant do?

Memory cue: Retrieve, answer, check each claim, and abstain without evidence.

## Guided laboratory

Run a small extractive retrieval and answer pipeline.

1. Search for a relevant policy
2. Produce an extractive classroom answer with a source ID
3. Record supported and unsupported questions
4. Compare top-1 and top-2 on practice cases

Individual change: Replace a relevant passage with an irrelevant one and observe abstention.

Expected evidence: Twenty-case practice table; ten final questions remain sealed for assessment.

## Checks and common error

Measure: Supported claims / checked claims and no-answer accuracy. Record the input, setting, source version, result and the relevant numerator and denominator.

Plausible incorrect claim: A citation label proves that the cited passage supports the answer.

Correction: Read the passage and check that it entails the claim; a plausible-looking source tag can be wrong.

If learners disagree, reproduce the same case and compare the underlying source or expected answer. Record unresolved cases for instructor review.

## Further discussion

- How could the conclusion change if the source or input changes?
- Which part of the result is observed and which part is an assumption?
- What simple test would detect the failure described above?

## References and optional extension

Core readings are official vendor documentation or original research. Read the relevant section; an account or paid API is unnecessary for this notebook. Page details may change before 2027.

Microsoft, Retrieval augmented generation.

<https://learn.microsoft.com/en-us/azure/foundry/concepts/retrieval-augmented-generation>

OpenAI, Evaluation best practices. <https://developers.openai.com/api/docs/guides/evaluation-best-practices>

Optional extension: replace one classroom proxy with a locally approved model or service, keep the original baseline, repeat the same checks, and state the additional privacy and cost assumptions.

Curriculum scope: the supplied 12-week teaching sequence, recorded in docs/SYLLABUS\_ALIGNMENT.md. The original module descriptor is retained in sources/originals/07-Descriptor-Generative-AI-new-.docx. Supporting project notes: Deep-research.txt, RAG.txt, AI-Agents.txt, MCP-&-Vector-Database.txt, Multi-Agent-AI-System.txt, Eval-&-Finetune&-Loop-Engineering.txt. These are planning sources, not empirical performance evidence.

## Student revision and exam preparation

Outcomes: LO4, LO5 | 2 hours concepts + 2 hours guided practice

### The idea in plain language

RAG combines retrieval with answer construction. Evaluate retrieval separately from answer support. A source label is not evidence unless its passage supports the claim. Tune settings on 20 practice questions. Freeze the choice before using the ten separate final questions.

### A worked example

For a question with two relevant passages, top-1 retrieves one and top-2 retrieves both. Recall rises from 1/2 to 2/2. This does not by itself show that generated claims are correct. A separate citation check is required.

### Practice before reading the answers

1. Why reserve final questions? (4 suggested practice marks)
2. Two of four material claims are supported. What is citation support? (4 suggested practice marks)
3. When should the assistant abstain? (4 suggested practice marks)

Draw the main diagram from memory, explain each arrow, then check it against the slides. Answer the three questions without notes before reading the model answers.

## Model answers and marking guidance

1. They estimate performance on cases that did not guide tuning. Repeated tuning on them leaks evaluation information.
2.  $2/4 = 50\%$ . Check each claim against its cited evidence.
3. When the approved evidence does not support an answer, including missing or irrelevant sources.

For each four-mark practice answer: two marks for the correct idea, one for a relevant example or calculation, one for an appropriate explanation or limitation. These suggested practice marks do not change the module assessment scheme.

## Official reading and video support

M1 Microsoft Learn: Introduction to retrieval-augmented generation concepts

<https://learn.microsoft.com/en-us/training/modules/rag-fundamentals/>

Study focus: Preparation, retrieval, generation and evaluation.

M3 Microsoft Learn: Building a RAG-powered AI chat app with Python and VS Code

<https://learn.microsoft.com/en-us/shows/vs-code-day-2024/building-a-rag-powered-ai-chat-app-with-python-and-vs-code>

Study focus: Optional system walkthrough. Older SDK interfaces are not the core lab dependency.

O1 OpenAI: Evaluation best practices

<https://developers.openai.com/api/docs/guides/evaluation-best-practices>

Study focus: Task-specific testing, repeated outputs and separate evaluation cases.

Selected source concepts inform the lesson. Examples, figures, questions and solutions are original classroom material; no exam questions are claimed to be official.